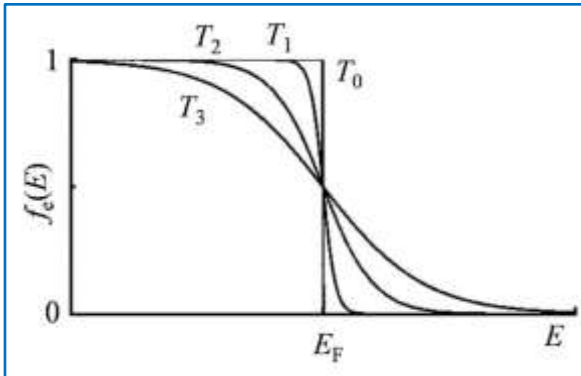


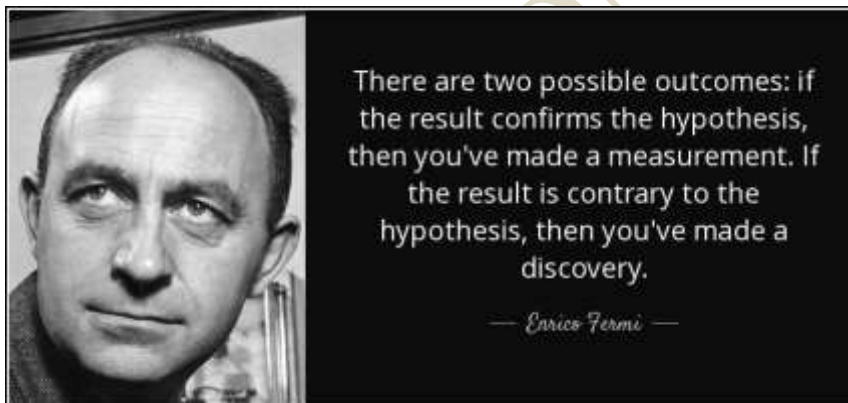


MODULE – 3

Electrical and Thermal Properties of Materials for MECHANICAL CLUSTER

Reference Books

1. Engineering Physics - Gaur and Gupta
2. A Text book of Engineering Physics - M.N. Avadhanulu and P.G. Kshirsagar
3. Solid State Physics - S O Pillai, 8th Ed - New Age International Publishers- 2018.



SYLLABUS

MODULE – 3	ELECTRICAL AND THERMAL PROPERTIES OF MATERIALS	[8 HOURS]
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Electrical Properties: Review of classical free electron theory, limitations of classical free electron theory. Postulates of quantum free electron theory, Fermi energy, Fermi velocity, Fermi temperature. Fermi factor and its dependence on energy and temperature. Electrical conductivity (qualitative expression using effective mass and Fermi velocity). Merits of quantum free electron theory. Problems.

Thermal Properties: Thermal conductivity, expression for thermal conductivity of a conductor using classical free electron theory. Wiedemann–Franz law, calculation of Lorentz number using classical and quantum assumptions. Theory and determination of thermal conductivity using Forbe’s and Lee–Charlton’s methods. Problems.

Practical Topics: 1. Fermi energy of copper, 2. Thermal conductivity of a poor conductor by Lee–Charlton’s method, and 3. Thermal conductivity of a good conductor by Forbe’s method

Self-study: Classical free electron theory

Classical free electron theory (CFET)

The first theory was developed by Drude and Lorentz in 1900. According to this theory, metal contains free electrons which are responsible for the electrical conductivity and electrons obey the laws of classical mechanics.

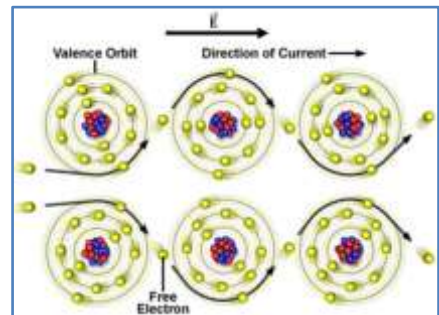
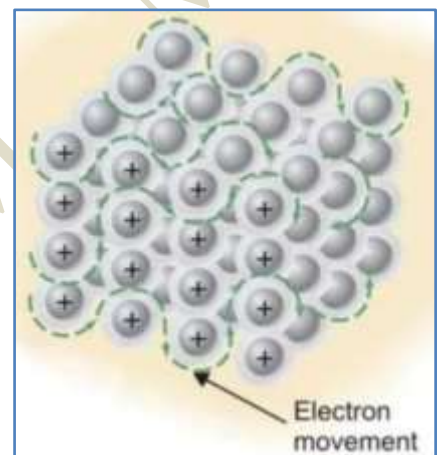
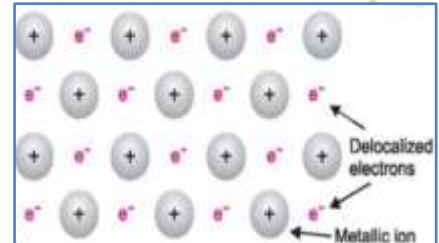
Assumptions/Postulates of the CFET

1. A metal is imagined as the structure of three dimensional arrays of atoms consisting of valence electrons.
2. Collection of valence electrons from all the atoms is free to move about the whole volume of the metals like the molecules of a perfect gas in a container.
3. The free electrons are treated as equivalent to gas molecules and they are assumed to obey the laws of kinetic theory of gases.
4. In metals, the positive ion cores are at fixed positions and the free electrons move in random directions and collide with either positive ions or other free electrons. All such collisions are considered elastic i.e., there is no loss of energy.
5. In the absence of the electric field, the energy associated with each electron at a temperature T is given by

$$KE = \frac{3}{2} k_b T \Rightarrow \frac{1}{2} m v_{th}^2 = \frac{3}{2} k_b T$$

where v_{th} is the thermal velocity, k_b is a Boltzmann constant (1.38×10^{-23} J/K).

6. The electric field produced due to the ionic cores is considered to be uniform throughout the body of the metal and it is neglected.
7. Force of repulsion between the electrons and force of attraction between electrons and lattice ions is neglected.
8. When an electric field is applied to the metal, the free electrons are accelerated in the direction opposite to the direction of applied electric field.



Note:

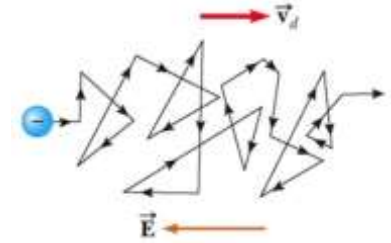
As we know that, the atomic packing fraction (APF), [APF is a measure of total volume of an unit cell occupied by the atoms] for

Unit cell	APF%	Free Space%
Simple cubic	52%	48%
Body Centered Cubic	68%	32%
Face Centered Cubic	74%	26%

Since, metals have large percentage of free space in them, it is reasonable to imagine that the valence electrons form an electron gas which is free to move about the whole volume of the metals like the molecules of a perfect gas in a container.

Drift Velocity (v_d): (This will not be asked in the exam)

Constant velocity acquired by the electrons under steady state in the presence of applied electric field is called Drift velocity.



Lorentz force experienced by the electrons in the presence of electric field is $F_L = eE$

Opposing force experienced by them due to the collision with the ionic cores is $F_r = \frac{mv_d}{\tau}$

When a system is in steady state, $F_r = F_L$

$$\frac{mv_d}{\tau} = eE$$

$$v_d = \frac{eE\tau}{m}$$

Mean free path (λ): (This will not be asked in the exam)

The average distance travelled by the conduction electrons between two successive collisions with lattice ions.

Mean collision time (τ): (This will not be asked in the exam)

The average time that elapses between two consecutive collisions of an electron with the lattice points is called mean collision time.

$$\tau = \frac{\lambda}{v}$$

where, λ is the mean free path and v is the thermal velocity (v_{th}) in the absence of electric field & drift velocity (v_d) in the presence of electric field.

Mean relaxation time (τ_r): (This will not be asked in the exam)

From the instant of sudden disappearance of an electric field across a metal, the average velocity of the conduction electrons decays exponentially to zero, and the time required in this process for the average velocity to reduce to $(1/e)$ times its value is known as Relaxation time.

Or

The time required for the drift velocity to decay to $(1/e)$ times its original value.

$$v_d = v_0 e^{\frac{-\tau}{\tau_r}}$$

where v_0 is the initial velocity of an electron in the absence of electric field and τ_r is the relaxation time.

if $\tau = \tau_r$ then $v_d = v_0 e^{-1}$ or $v_d = \frac{v_0}{e}$

Microscopic form of Ohm's law

We know that,

$$V = IR \text{ and } R = \frac{\rho L}{A}$$

$$V = I \left(\frac{\rho L}{A} \right)$$

$$\frac{I}{A} = \frac{1}{\rho} \times \frac{V}{L}$$

$$J = \sigma E$$

V	- Voltage (V)
I	- Current (A)
R	- Resistance (Ω)
ρ	- Resistivity (Ωm)
L	- Length of the conductor (m)
A	- Area of the conductor (m^2)
J	- Current density (A/m^2)
σ	- Conductivity ($\Omega\text{m})^{-1}$
E	- Electric field (V/m)

Expression for Electrical Conductivity in Metals (This will not be asked in the exam)

- Consider a conductor of uniform area of cross section A, length L, carrying a current I when an electric field E is applied.
- Let **n** be the free electron density (i.e, number of free electrons per unit volume in a conductor)

- Total Number of electrons (**N**) in the metallic conductor is

$$N = \frac{\text{electrons}}{\text{unit volume}} \times \text{Total volume of the conductor}$$

$$N = n \times Al$$

- If **e** is the charge of the electron, then Net charge (**Q**) flowing in the conductor

$$Q = eN$$

$$Q = enAl$$

- Current (**I**) flowing through the conductor is given by

$$I = \frac{Q}{t}$$

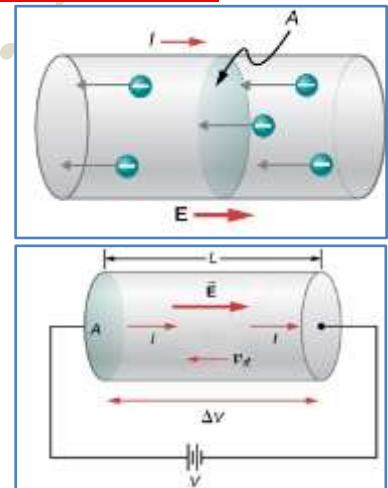
$$I = \frac{enAl}{t} = enA \left(\frac{l}{t} \right) = enAv_d$$

$$I = enAv_d \text{ ---- (1)}$$

- wkt, $\vec{J} = \sigma \vec{E}$ (microscopic form of ohm's law)

$$\sigma = \frac{J}{E} = \frac{I}{AE} \text{ ---- (2)}$$

Substituting equation (1) in (2)



$$\sigma = \frac{enAv_d}{AE} = \frac{env_d}{E}$$

$$wkt, v_d = \frac{e E \tau}{m}$$

$$\sigma = \frac{en}{E} \left(\frac{e E \tau}{m} \right)$$

$$\sigma = \frac{ne^2\tau}{m}$$

where, σ is conductivity, n is electron density, e is charge of electron, τ is mean collision time, and m is mass of electron

$$\Rightarrow \sigma \propto n$$

“Electrical conductivity of metals is directly proportional to electron density”

Electron mobility (μ) (This will not be asked in the exam)

It is defined as the drift velocity acquired by an electron per unit of electric field strength.

$$\mu = \frac{\text{drift velocity}}{\text{unit electric field}} = \frac{v_d}{E}$$

$$\sigma = \frac{env_d}{E} = en\mu$$

$$\sigma = en\mu$$

$$\sigma \propto \mu$$

“Electrical conductivity of metals is directly proportional to mobility of electron”

Expression for σ in terms of temperature (This will not be asked in the exam)

we know that,

$$\sigma = \frac{ne^2\tau}{m}$$

according to CFET, Kinetic energy of the electrons is given by

$$\frac{1}{2}mv_{th}^2 = \frac{3}{2}k_bT$$

$$v_{th} = \sqrt{\frac{3k_bT}{m}}$$

$$\text{but, } \tau = \frac{\lambda}{v_{th}}$$

therefore,

$$\tau = \frac{\lambda\sqrt{m}}{\sqrt{\frac{3k_bT}{m}}} = \frac{\lambda\sqrt{m}}{\sqrt{3k_bT}}$$

therefore,

$$\sigma = \frac{ne^2\lambda}{\sqrt{3mk_bT}}$$

$$\Rightarrow \sigma \propto \frac{1}{\sqrt{T}}$$

“Electrical conductivity of metals is inversely proportional to square root of the temperature”

Successes of the CFET

CFET successfully explains

- i. Ohm's law
- ii. bulk electrical and thermal conductivity of metals.
- iii. Wiedemann-Franz Law.
- iv. Optical properties of metals

Failures of CFET

CFET fails to account for the experimental results obtained for the properties such as

- i. Specific heat at constant volume (C_V)
- ii. Dependence of electrical conductivity on temperature
- iii. Dependence of electrical conductivity on electron concentration

i. Specific heat constant volume (C_V)

- Total specific heat for a material is the sum of contributions from electrons and thermal vibrations i.e., $C_V = C_{electron} + C_{Phonon}$
- According to CFET,

$$E = \frac{3}{2}k_bT \Rightarrow \frac{dE}{dT} = \frac{3}{2}k_b$$

For all free electrons (n) per unit volume in the metal

$$C_{electron} = \frac{3}{2}nk_b = 1.5R$$

- But, according to experiment, $C_{electron} = 10^{-4}RT$
- Experimental value is very much less than the theoretical value.
- Also according to CFET, the specific heat is independent of temperature but experimentally it depends on temperature

“Hence the classical free electron theory fails to explain the specific heat of metals”

ii. Dependence of electrical conductivity (σ) on temperature (T)

- **Theoretically** - according to the assumptions of classical theory that the kinetic energy is given by: $\frac{3}{2}k_bT = \frac{1}{2}mv_{th}^2$, where v_{th} is the thermal velocity.

$$\sigma = \frac{ne^2\lambda}{\sqrt{3mk_bT}}$$

$$\sigma_{Th} \propto \frac{1}{\sqrt{T}}$$

- **Experimentally**

$$\sigma_{exp} \propto \frac{1}{T}$$

“Hence the classical free electron theory fails to explain the dependence of σ on T ”

iii. Dependence of electrical conductivity (σ) on electron concentration (n)

- **Theoretically** –

$$\sigma = \frac{ne^2\tau}{m}$$

$$\Rightarrow \sigma \propto n$$

According to the above relation, the electrical conductivity of divalent metals like zinc and cadmium must be higher compared to that of monovalent metals like copper and silver.

- **Experimentally** –

But, according to experimental results, the monovalent copper and silver metals have high conductivity compared to that of divalent Cadmium and Zinc, as given in the below table.

“Hence the classical free electron theory fails to explain the dependence of σ on n ”

Valence state	Metal	Electron density (n)	Electrical conductivity (σ)
Monovalent	Copper	$8.45 \times 10^{28}/m^3$	$5.88 \times 10^7/\Omega m$
	Silver	$5.85 \times 10^{28}/m^3$	$6.3 \times 10^7/\Omega m$
Divalent	Cadmium	$9.28 \times 10^{28}/m^3$	$0.15 \times 10^7/\Omega m$
	Zinc	$13.1 \times 10^{28}/m^3$	$1.09 \times 10^7/\Omega m$

Quantum free electron theory (QFET)

Quantum free electron theory was proposed by Arnold Sommerfeld in 1928. It overcomes many of the drawbacks of classical theory. Sommerfeld explained them by choosing Fermi Dirac statistics instead of Maxwell-Boltzmann statistics. He developed this theory by applying the principles of quantum mechanics.

Assumptions/Postulates of the QFET

1. Electrons are tiny, indistinguishable particles called **fermions** having half integral spin.
2. The velocity and energy distribution of the free electrons are governed by the **Fermi-Dirac distribution function**.
3. The allowed energy levels of electrons are quantized.
4. The distribution of electrons in the various allowed energy levels obey **Pauli's exclusion principle**.
5. The electric field produced due to the ionic cores is considered to be uniform throughout the body of the metal and it is neglected.
6. Force of repulsion between the electrons and force of attraction between electrons and lattice ions is neglected.

Density of states, $g(E)$

It gives the distribution of all the possible quantum states per unit volume per unit energy that the electrons in the metal can take.

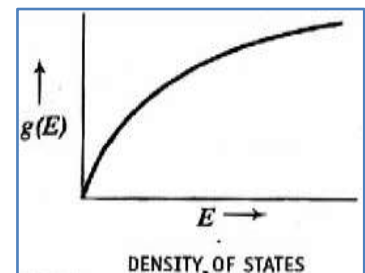
$$g(E) = \frac{\text{number of allowed energy levels per unit energy interval}}{\text{unit volume of the material}}$$

$$g(E)dE = \frac{8\sqrt{2}\pi m^{\frac{3}{2}}}{h^3} E^{\frac{1}{2}} dE$$

$$g(E)dE \propto E^{\frac{1}{2}} dE$$

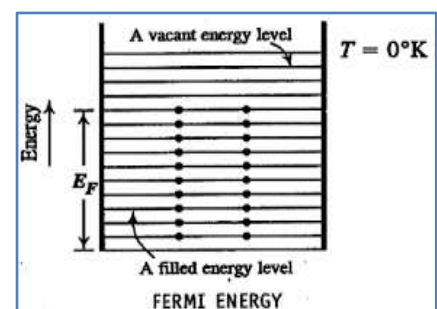
Integration of this function between any two energy values gives the total number of quantum states per unit volume that exist between those energy values

$$\int_{E_1}^{E_2} g(E)dE = \frac{8\sqrt{2}\pi m^{\frac{3}{2}}}{h^3} \int_{E_1}^{E_2} E^{\frac{1}{2}} dE$$



Fermi energy and Fermi energy level

- In a metal having N atoms, there are N allowed energy levels in each band. In the energy band the energy levels are separated by energy differences.
- According to Pauli's exclusion principle, each energy level can accommodate a maximum of two electrons, one with spin up



and the other with spin down.

- The filling up of energy levels occurs from the lowest level. The next pair of electrons occupies the next energy level and so on till all the electrons in the metal are accommodated.
- **The energy of the highest occupied level at absolute zero temperature (0K) is called the Fermi energy. And the corresponding energy level is called Fermi level denoted by (E_F)**
- At 0K and absence of external electric field, all the levels above the Fermi level are empty and those lying below Fermi level are completely filled.

Fermi velocity (v_F)

The maximum velocity of the electrons which are present at the Fermi level is called Fermi velocity.

$$E_F = \frac{1}{2} m v_F^2$$

$$v_F = \left(\frac{2E_F}{m} \right)^{\frac{1}{2}}$$

where, v_F is Fermi velocity, E_F is Fermi energy, and m is mass of the electron.

Explanation (This will not be asked in the exam)

When the velocities of the valence electrons are plotted as vectors in space, the end points of the vector correspond to a sphere, called the Fermi surface. Fermi velocity (v_F) is the maximum velocity of the electron having highest energy. The Fermi surface has a radius equal to Fermi velocity v_F . In the absence of the field, the net velocity of fermions and hence the current through the metal is zero as the velocity vectors cancel out one another. When an electric field is applied, the electron movement with high velocity v_F results in current and the inner electrons movement with lesser velocity do not contribute to current.

Fermi Surface (This will not be asked in the exam)

If we plot the velocities of all the free electrons in velocity space, such that the end points of velocity exist inside a surface, it will be spherical with the radius equal to the maximum velocity (Fermi velocity). This spherical surface is termed Fermi Surface. Current in the metal is due to motion of free electrons moving with almost maximum velocity, v_F , which are very near the periphery of Fermi surface.

Fermi temperature (T_F)

The temperature at which average kinetic energy of an electron in a solid is equal to the thermal energy at 0 K. But the thermal energy possessed by electrons is given by the product kT .

When $T = T_F$, the equation

$$E_F = kT_F$$

$$T_F = \frac{E_F}{k}$$

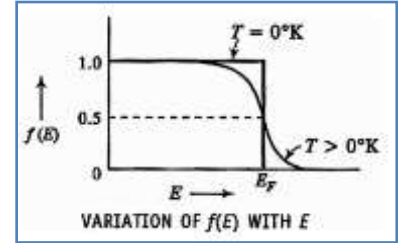
where, T_F is Fermi temperature, E_F is Fermi energy, and k is Boltzmann constant.

Fermi function, $f(E)$

It gives the probability of occupation of a given energy state by electrons for a material in thermal equilibrium.

The Fermi function, $f(E)$ at any given temperature, T is given by,

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}$$

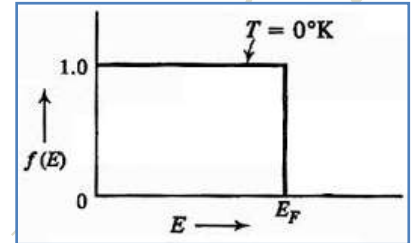


Effect of temperature and energy on Fermi function

Case 1: for $E < E_F$ and $T = 0 K$

$$f(E) = \frac{1}{e^{-\infty} + 1} = \frac{1}{0 + 1} = 1$$

$$\therefore f(E) = 1 \text{ for } E < E_F \text{ and } T = 0 K$$



“Hence all the energy levels below Fermi level are completely filled at 0 K”

Case 2: for $E > E_F$ and $T = 0 K$

$$f(E) = \frac{1}{e^{\infty} + 1} = \frac{1}{\infty} = 0$$

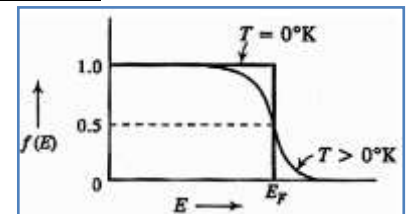
$$\therefore f(E) = 0 \text{ for } E > E_F \text{ and } T = 0 K$$

“Hence all the energy levels above Fermi level are completely empty at 0 K”

Case 3: for $E = E_F$ and $T > 0 K$

$$f(E) = \frac{1}{e^0 + 1} = \frac{1}{1 + 1} = \frac{1}{2}$$

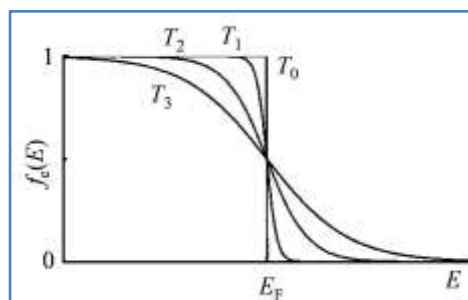
$$\therefore f(E) = \frac{1}{2} \text{ for } E = E_F \text{ and } T > 0 K$$



“Hence, as the temperature increases, the electrons close to the Fermi level shifts to higher energy

levels which were empty at 0 K”

Variation of $f(E)$ at different temperatures ($T_0 < T_1 < T_2 < T_3$)

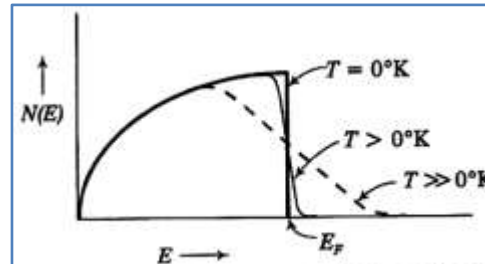


Electron density of states, $N(E)$

- The number of electrons with energy range E & $(E+dE)$ in unit volume is $N(E) dE$
- $N(E) dE$ is given by the product of the number of energy levels in the same range and the Fermi factor.

$$N(E)dE = f(E) \times g(E)dE$$

- The plot of $N(E)dE$ versus E represents the actual distribution of electrons among the available states for the material at different temperatures.



Electron density, n

The number of electrons per unit volume of the material ' n ' can be evaluated by integrating the above expression from $E = 0$ to $E = E_{max}$, where E_{max} is the maximum energy possessed by the electrons.

$$n = \int_{E=0}^{E=E_{max}} N(E) dE$$

$$n = \int_{E=0}^{E=E_{max}} g(E) f(E) dE$$

Expression for electron density, n and Fermi energy, E_F at 0 K

we know that,

$$\text{At } T = 0K, f(E) = 1 \text{ and } E_{max} = E_{F0}$$

we know that,

$$g(E)dE = \frac{8\sqrt{2}\pi m^{\frac{3}{2}}}{h^3} E^{\frac{1}{2}} dE$$

therefore,

$$n = \int_{E=0}^{E=E_{max}} g(E) f(E) dE$$

$$n = \frac{8\sqrt{2}\pi m^{\frac{3}{2}}}{h^3} \int_0^{E_{F0}} E^{\frac{1}{2}} \times 1 dE$$

$$n = \frac{16\sqrt{2}\pi m^{\frac{3}{2}}}{3h^3} (E_{F0})^{\frac{3}{2}}$$

$$(E_{F0})^{\frac{3}{2}} = \frac{3h^3 n}{16\sqrt{2}\pi m^2}$$

$$E_{F0} = \left(\frac{h^2}{8m}\right) \left(\frac{3n}{\pi}\right)^{\frac{2}{3}} \Rightarrow \left(\frac{h^2}{8m}\right) \left(\frac{3}{\pi}\right)^{\frac{2}{3}} (n)^{\frac{2}{3}}$$

$$E_{F0} = B (n)^{\frac{2}{3}}$$

where, $B = \left(\frac{h^2}{8m}\right) \left(\frac{3}{\pi}\right)^{\frac{2}{3}} = 5.85 \times 10^{-38} J$

Quantum expression for electrical conductivity

$$\sigma = \frac{ne^2\tau}{m^*} = \frac{ne^2\lambda}{m^*v_F}$$

where, m^* is effective mass of electron, λ is mean free path, and v_F is Fermi velocity

Successes of QFET

- Specific heat at constant volume (C_V)
- Dependence of electrical conductivity on temperature
- Dependence of electrical conductivity on electron concentration

i. Specific heat constant volume (C_V)

- According to CFET, $C_{electron} = \frac{3}{2}nk_b = 1.5R$
- According to QFET, the expression for specific heat is given by

$$C_V = \left(\frac{2k}{E_F}\right) RT$$

Taking $E_F \approx 5eV$ for metals, therefore we get

$$\frac{2k}{E_F} \approx 10^{-4}$$

$$\therefore C_V = 10^{-4}RT$$

This value is in excellent agreement with the experimental value.

“Hence, the QFET successfully explain the specific heat property of metals”

ii. Dependence of electrical conductivity (σ) on temperature (T)

- According to the CFET,

$$\sigma_{Th} \propto \frac{1}{\sqrt{T}}$$

- From the quantum expression, $= \frac{ne^2\lambda}{m^*v_F}$, $\sigma \propto \lambda$ which is inversely proportional to temperature T. So, conductivity is inversely proportional to T, as verified by experiments.

- According to the QFET,

$$\sigma_{Th} \propto \frac{1}{T}$$

“Hence the QFET successfully explain the dependence of σ on T ”

iii. Dependence of electrical conductivity (σ) on electron concentration (n)

- According to the CFET, $\sigma = \frac{ne^2\tau}{m} \Rightarrow \sigma \propto n$

According to the above relation, the electrical conductivity of divalent metals like zinc and cadmium must be higher compared to that of monovalent metals like copper and silver. But, the mono valent copper and silver has higher conductivity compared to that of the divalent zinc and cadmium.

- According to QFET, $\sigma = \left(\frac{ne^2}{m^*}\right) \frac{\lambda}{v_F}$

From the above equation, it is clear that, σ depends both on the n and $\frac{\lambda}{v_F}$. Therefore, though the value of ‘n’ for divalent metals is higher than monovalent metals, the $\frac{\lambda}{v_F}$ will be material specific value which affects the overall conductivity of the given metal. For example, the value of n for aluminium is 2.13 times higher than that of copper. But the value of $\frac{\lambda}{v_F}$ for copper is about 3.73 times higher than that of aluminium. Thus the conductivity of copper exceeds that of aluminium.

“Hence the QFET successfully explain the dependence of σ on n ”

Problems:

Formula	
Current density	$J = \frac{I}{A} = \frac{neAv_d}{A} = nev_d$
Drift velocity	$v_d = \frac{eE\tau}{m}$
Electrical Conductivity	$\sigma = \frac{ne^2\tau}{m}$
Density of states	$g(E)dE = \frac{8\sqrt{2}\pi m^{\frac{3}{2}}}{h^3} E^{\frac{1}{2}} dE$
Fermi energy	$E_{F0} = \left(\frac{h^2}{8m}\right) \left(\frac{3}{\pi}\right)^{\frac{2}{3}} (n)^{\frac{2}{3}} = B (n)^{\frac{2}{3}}$ where, $B = \left(\frac{h^2}{8m}\right) \left(\frac{3}{\pi}\right)^{\frac{2}{3}} = 5.85 \times 10^{-38} \text{ J}$
Fermi function	$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}$
Fermi velocity	$v_F = \left(\frac{2E_F}{m}\right)^{\frac{1}{2}}$
Fermi temperature	$T_F = \frac{E_F}{k}$



P1. Calculate the Fermi energy of sodium assuming that the metal has one free electron per atom. Given the density of sodium = 970 kg/m³ and atomic weight of sodium = 22.99.

Soln:

$$E_{F0} = \left(\frac{h^2}{8m} \right) \left(\frac{3}{\pi} \right)^{\frac{2}{3}} (n)^{\frac{2}{3}}$$

$$n = \frac{\text{No. of valence electron} \times N_A \times \text{Density}}{\text{Atomic weight}}$$

$$n = \frac{1 \times 6.023 \times 10^{26} \times 970}{22.99} = 2.54 \times 10^{28}$$

$$E_{F0} = 3.92 \text{ eV}$$

P2. Determine the drift velocity of electrons in a copper conductor with cross sectional area of 10⁻⁶ m² carrying a current of 4 A. The atomic weight of Cu is 63.6 and density is 8.9 g/cm³.

Soln:

$$n = \frac{\text{No. of valence electron} \times N_A \times \text{Density}}{\text{Atomic weight}}$$

$$= 8.424 \times 10^{28}$$

$$I = neAv_d$$

$$v_d = \frac{I}{neA} = 3 \times 10^{-4} \text{ m/s}$$

P3. Determine the relaxation time in a current carrying metallic wire. The metal has 5.8 × 10²⁸ conduction electrons/m³ and its resistivity is 1.54 × 10⁻⁸ Ωm.

Soln:

$$\sigma = \frac{ne^2}{m}$$

$$\tau = \frac{m}{ne^2} \sigma = \frac{m}{ne^2} \frac{1}{\rho} = 3.97 \times 10^{-14} \text{ s}$$

P4. Calculate the probability of an electron occupying a energy level 0.02 eV above the

Fermi level at 200 K and 400 K in a material.

Soln:

$$E - E_F = 0.02 \text{ eV} = 3.2 \times 10^{-21} \text{ J}$$

$$\text{i. At 200K, } f(E) = 0.24 = 24\%$$

$$\text{ii. At 400K, } f(E) = 0.36 = 36\%$$

P5. Calculate the probability of an electron occupying an energy level 0.02 eV above and below the Fermi level at 200 K.

Soln:

$$\text{i. } E - E_F = 0.02 \text{ eV} = 3.2 \times 10^{-21} \text{ J}$$

$$\text{At 200K, } f(E) = 0.24 = 24\%$$

$$\text{ii. } E - E_F = 0.02 \text{ eV} = -3.2 \times 10^{-21} \text{ J}$$

$$\text{At 200K, } f(E) = 0.76 = 76\%$$

P6. Find the temperature at which there is 1% probability that a state with an energy 0.5 eV above Fermi energy is occupied.

Soln:

$$f(E) = \frac{1}{100} \text{ and } E - E_F = 0.5 \text{ eV} = 8 \times 10^{-20} \text{ J}$$

$$\frac{1}{100} = \frac{1}{e^{\frac{8 \times 10^{-20}}{1.38 \times 10^{-23} T}} + 1}$$

$$e^{\frac{8 \times 10^{-20}}{1.38 \times 10^{-23} T}} + 1 = 100$$

$$e^{\frac{5.797 \times 10^3}{T}} = 100 - 1$$

$$\frac{5.797 \times 10^3}{T} = \ln(99)$$

$$T = 1261.58 \text{ K}$$

P7. Consider the Fermi energy of copper at 8 eV. Calculate the Fermi velocity of free electrons in copper.

Soln:

$$v_F = \left(\frac{2E_F}{m} \right)^{\frac{1}{2}} = 1.676 \times 10^6 \text{ m/s}$$

**Thermal Properties of solids****Thermal Conductivity (κ)**

Thermal conductivity of a material is equal to the amount of heat energy (Q) conducted per unit area of cross section to the temperature gradient $\left(\frac{dT}{dx}\right)$.

The amount of heat energy conducted per unit area of cross section per unit time is directly proportional to the temperature gradient.

i.e.,

$$Q \propto \frac{dT}{dx}$$

$$Q = \kappa \frac{dT}{dx}$$

$$\kappa = \frac{Q}{dT/dx}$$

The thermal conductivity is produced due to the presence of both free electrons and thermally excited lattice vibrations known as phonons.

$$\kappa_{total} = \kappa_{electrons} + \kappa_{phonons}$$

In case of metals, since the electron concentration is very high, the thermal conductivity is only due to the presence of electrons.

For insulators, the contribution to the thermal conductivity is mainly from the lattice vibrations, as there will be no valence electrons in insulator.

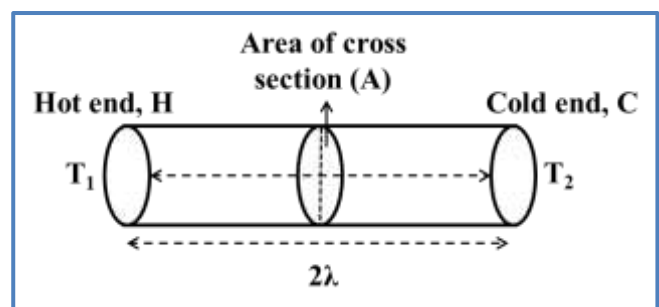
Expression for thermal conductivity

The thermal energy transfer is always from high temperature medium to the low temperature medium, and the energy transfer stops when the two mediums are at the same temperature (equilibrium state).

Let us consider a uniform metal rod, HC with temperature at the hot end as ' T_1 ' and temperature at the cold end ' T_2 '.

Let ' A ' be the area of cross section and ' 2λ ' be the distance between the two ends.

Let ' Q ' be the amount of heat energy flowing through the conductor



$$Q \propto \frac{A(T_1 - T_2)t}{2\lambda}$$



$$Q = \frac{\kappa A (T_1 - T_2) t}{2\lambda}$$

$$\kappa = \frac{Q 2\lambda}{A (T_1 - T_2) t}$$

For thermal conductivity per unit area ($A = 1 \text{ unit}$) per unit time ($t = 1 \text{ unit}$).

$$\kappa = \frac{Q 2\lambda}{(T_1 - T_2)}$$

Let 'n' be the number of conduction electrons and 'v' be the root mean square velocity of electrons.

Let us assume that the available free electrons in metal are having equal probability to move in all possible six directions. Therefore, an average of ' $\frac{1}{6}nv$ ' electrons can travel in one direction.

According to classical free electron theory, the free electrons in metal behave like molecules of gas. Therefore, the average kinetic energy of electrons at the hot end is

$$T_1 = \frac{3}{2} k_B T_1 \text{ --- (2)}$$

Similarly, the average kinetic energy of electrons at the cold end is

$$T_2 = \frac{3}{2} k_B T_2 \text{ --- (3)}$$

The heat energy conducted from hot end to cold end is equal to the product of number of electrons and the average KE of electrons, i.e.,

$$Q = \frac{1}{6} nv \times \frac{3}{2} k_B T_1$$

$$Q = \frac{1}{4} nv k_B T_1 \text{ --- (4)}$$

similarly, the heat energy conducted from cold end to hot end is equal

$$Q = \frac{1}{4} nv k_B T_2 \text{ --- (5)}$$

total heat energy conducted is given by

$$Q = \frac{1}{4} nv k_B (T_1 - T_2) \text{ --- (6)}$$

the temperature gradient is given by

$$\frac{dT}{dx} = \left(\frac{T_1 - T_2}{2\lambda} \right) \text{ --- (7)}$$

Dividing equation (6) by (7), we get



$$\frac{Q}{dT/dx} = \frac{\frac{1}{4}nvk_B(T_1 - T_2)}{\frac{T_1 - T_2}{2\lambda}}$$

$$\kappa = \frac{1}{2}nvk_B\lambda \text{ (W/m/K)} \text{ --- (8)}$$

we know that, from CFET, the electronic heat capacity of the metal is,

$$C_e = \frac{3}{2}R = \frac{3}{2}nk_B$$

$$nk_B = \frac{2}{3}C_e$$

$$(8) \Rightarrow \kappa = \frac{1}{2}v\lambda \left(\frac{2}{3}C_e \right)$$

$$\therefore \kappa = \frac{1}{3}C_e v\lambda \text{ (W/m/K)} \text{ --- (9)}$$

The thermal conductivity of a metal is determined either using equation (8) or (9).

Weidemann-Franz law

Statement: “*The ratio of thermal conductivity to electrical conductivity of a metal is directly proportional to absolute temperature*”.

i.e.,

$$\frac{\kappa}{\sigma} \propto T$$

$$\frac{\kappa}{\sigma} = LT$$

where, L is Lorentz number ($L = 2.44 \times 10^{-8} W\Omega K^{-2}$)

Expression for Lorentz number:

We know that, thermal conductivity, $\kappa = \frac{1}{2}nvk_B\lambda \text{ --- (1)}$

We know that, electrical conductivity, $\sigma = \frac{ne^2\tau}{m} \text{ --- (2)}$

therefore,

$$\frac{\kappa}{\sigma} = \frac{\frac{1}{2}nvk_B\lambda}{\frac{ne^2\tau}{m}} = \frac{mvk_B\lambda}{2e^2\tau} = \frac{1}{2} \frac{mvk_B\lambda}{e^2\tau}$$

we know that,

$$\lambda = v\tau$$



$$\frac{\kappa}{\sigma} = \frac{1}{2} \frac{m v k_B v \tau}{e^2 \tau}$$

$$\frac{\kappa}{\sigma} = \frac{1}{2} m v^2 \frac{k_B}{e^2}$$

but,

$$\frac{1}{2} m v^2 = \frac{3}{2} k_B T$$

$$\therefore \frac{\kappa}{\sigma} = \frac{3}{2} k_B T \times \frac{k_B}{e^2}$$

$$\frac{\kappa}{\sigma} = \frac{3 k_B^2}{2 e^2} T$$

$$\frac{\kappa}{\sigma} = L T$$

where,

$$L = \frac{3 k_B^2}{2 e^2} (W \Omega K^{-2})$$

Calculation of Lorentz number by CFET and QFET

(i) By CFET

$$\frac{\kappa}{\sigma} = L T$$

$$L = \frac{3 k_B^2}{2 e^2}$$

$$L = \frac{3 (1.38 \times 10^{-23})^2}{2 (1.6 \times 10^{-19})^2}$$

$$L = 1.115 \times 10^{-8} (W \Omega K^{-2})$$

Thus the value of Lorentz number determined from CFET is not in good agreement with the experimental value of $L = 2.44 \times 10^{-8} (W \Omega K^{-2})$. This is also one of the drawbacks of CFET.

(ii) By QFET

According to QFET, the thermal conductivity is given by

$$\kappa = \frac{n \pi^2 k_B^2 T \tau}{3 m}$$

we have,

$$\frac{\kappa}{\sigma} = L T$$



$$L = \frac{\kappa}{\sigma T}$$

$$L = \frac{\kappa}{\sigma T} = \frac{\frac{n\pi^2 k_B^2 T \tau}{3m}}{\frac{ne^2 \tau}{m} T} = \frac{\pi^2 k_B^2}{3 e^2}$$

$$L = \frac{\pi^2 k_B^2}{3 e^2}$$

$$L = \frac{3.14^2}{3} \times \frac{(1.38 \times 10^{-23})^2}{(1.6 \times 10^{-19})^2}$$

$$L = 2.44 \times 10^{-8} (W\Omega K^{-2})$$

Thus the value of Lorentz number determined from QFET is in excellent agreement with the experimental value of $L = 2.44 \times 10^{-8} (W\Omega K^{-2})$.

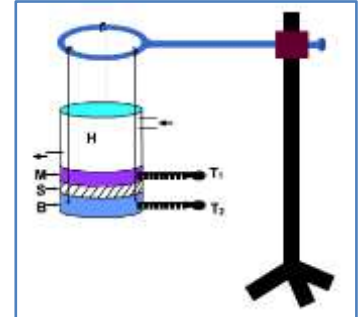
**Thermal conductivity of a Bad conductor by Lee's method**

AIM: To determine the thermal conductivity of bad conductors such as card board, ebonite, etc.

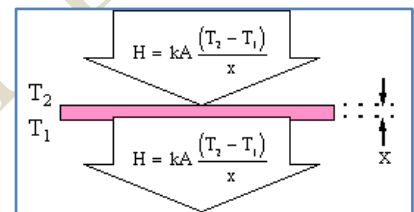
Apparatus: Lee's disc apparatus, a brass disc and a metallic disc of large diameter compared to its thickness, sample of bad conductor in the form of the disc with same diameter as the metallic disc, two thermometers, a stop watch, and a screw gauge.

Experimental Setup:

The experimental set up is shown in the figure. The brass disc B is hung from the stand with the help of three strings. On the brass disc, the bad conductor sample disc 'S' is placed and over that a metallic disc M is placed. On the metallic disc, a heating chamber, H with facility for steam-in and steam-out is placed.

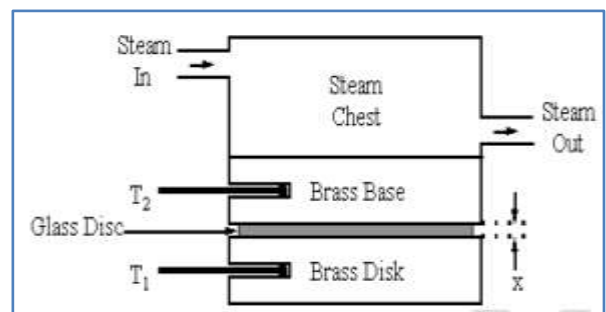
**Theory:**

Assuming that the heat losses to from the sides of the sample are negligible, the steady state rate of heat transfer (H) by conduction is given by: $H = \kappa A \frac{(T_2 - T_1)}{x}$, where the κ is the thermal conductivity of the sample, A is the cross sectional area, and $T_2 - T_1$ is the temperature difference across the sample thickness x as shown in the figure.

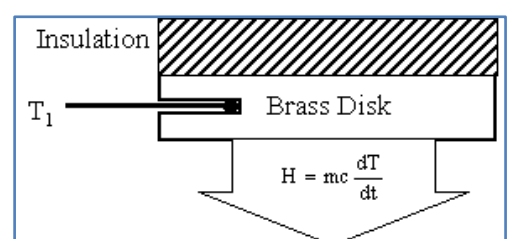


As the sample is an insulator, lagging at the sides will not significantly reduce the energy losses. Therefore to keep these losses small the sample is a thin disk with a large cross sectional area $\left[A = \frac{\pi d^2}{4}\right]$ compared to the area exposed at the edge $[a = x dx]$. Keeping A large and x small produces a large rate of energy transfer across the sample. Keeping x small also means that the apparatus reaches a steady state (when temperatures T_1 and T_2 are constant) more quickly.

The thin sample is sandwiched between the brass disc and brass base of the steam chest (see figure). The temperature of the brass base (measured by thermometer T_2) is very close to the temperature of the top surface of the bad conductor disk because the thermal conductivity of brass is about one hundred times that of bad conductor. Similarly the temperature of the brass disk (measured by T_1) is very close to the temperature of the lower bad conductor surface. In this way the temperature difference across such a thin sample can be accurately measured.



When the apparatus is in a steady state (temperatures T_1 and T_2 constant), the rate of heat conduction into the brass disk must be equal to the rate of heat loss due to cooling (by air convection) from the bottom of the brass disk. The rate of heat loss can be determined by measuring how fast the brass disk cools at the previous (steady state) temperature T_1 (with the





top of the brass disk covered with insulation, as shown in the figure. If the disk cools down at a rate $\frac{dT}{dt}$ then the rate of heat loss is given by: $H = mc \frac{dT}{dt}$, where m is the mass of the brass disk and c is the specific heat capacity of brass.

When a steady state is reached, let T_1 and T_2 be the temperatures of metallic disc M and brass disc B. So, the temperature difference between the two ends of bad conductor is $(T_1 - T_2)$.

Therefore, the rate of heat conducted through the bad conductor is,

$$Q_1 = \frac{\kappa A (T_2 - T_1)}{d} \quad \text{--- (1)}$$

where, 'd' is thickness of the bad conductor and 'A' is the area of cross section.

The rate of heat lost by the brass disc to the surrounding under steady state is,

$$Q_2 = mc \left(\frac{dT}{dt} \right)_{T_2} \quad \text{--- (2)}$$

Where m-mass of the brass disc, C is the heat capacity of brass disc and $\frac{dT}{dt}$ is its rate of cooling at T_2

From eqn (1) and (2),

$$\kappa = \frac{mcd \left(\frac{dT}{dt} \right)_{T_2}}{A(T_2 - T_1)} \quad \text{--- (3)}$$

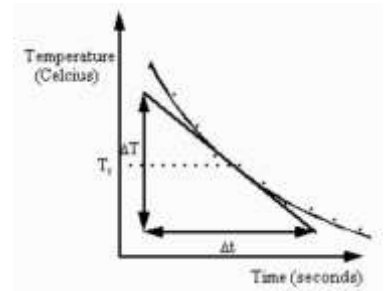
By measuring, $\left(\frac{dT}{dt} \right)_{T_2}$ and $(T_2 - T_1)$ using Lee's apparatus the thermal conductivity κ can be determined.

Procedure:

1. Measure the mass of the brass disc using a balance and take it as m kg.
2. Measure the diameter and hence the radius of the sample, S using vernier callipers. Calculate the circular area of the disc, $A = \pi r^2$.
3. Measure the thickness of the sample S using a screw gauge and take it as d.
4. Arrange the brass disc, sample disc and metallic disc as shown in the Figure 1. Insert the thermometers into the grooves of metallic disc and brass disc, which measure the temperatures T_1 and T_2 respectively.
5. Place the heating chamber on the above assembly and suspend it as shown in Figure 1. Start the heater, by sending steam through the heating chamber.
6. Monitor the temperatures T_1 and T_2 at a regular interval till they reach the steady state. Note the steady state temperatures T_1 and T_2 .
7. The next part is to determine the rate of cooling of brass disc. Lift the heating chamber and remove the metallic plate M and sample disc S, then place the heating chamber directly on the brass disc, B.
8. Allow the brass disc B to heat at least about 10°C above the steady state temperature T_2 measured in the first part of the experiment. Remove the heating chamber.



9. Switch on the stop watch and measure the temperature of brass disc at an interval of 60 sec as it cools down. Allow the brass disc to cool down to at least 5 0C below the steady state temperature, T_2 . The suggested table is as shown below.
10. Plot a graph of temperature T of brass disc as a function of time. A typical plot is shown in figure. Draw tangential line to the curve, corresponding to the temperature, T_2 and determine its slope. The slope is equivalent to $\left(\frac{dT}{dt}\right)_{T_2}$.



Rate of cooling of brass disc:

Sl. No.	Time (min)	Time in s	Temperature of steel disc T °C
1	0	0	
2	1	60	
3	2	120	
4	3	180	
5	4	240	
6	5	300	
7	6	360	
8	7	420	
9	8	480	
10	9	540	
11	10	600	

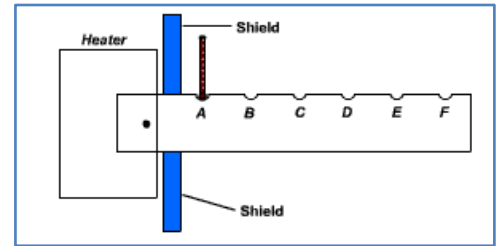
Result: Thermal conductivity of the given poor conductor specimen is found to be, $\kappa = \dots\dots\dots \text{W/mK}$

**THERMAL CONDUCTIVITY OF A METAL BY FORBE'S METHOD**

Aim: To measure the thermal conductivity of a good conductor by Forbe's method.

Apparatus: A long uniform hollow metal rod with holes drilled at appropriate places with semiconductor thermometers (six), temperature and time reading unit.

Experimental Setup: The experimental set up is shown in figure. One end of the experimental rod is connected to a Heating Chamber. There is a shield to avoid radiation and direct heating of other portion of the rod from the heater.



Formula: The coefficient of thermal conductivity of the metal rod is given by

$$k = \frac{\rho s \sum_0^L \frac{\Delta T}{\Delta t} \Delta x}{\left(\frac{\Delta T}{\Delta x}\right)_{x_2 \sim x_5}} \text{ W/mK}$$

where, ρ is the density of the metal rod in kg/m^3 , s is its specific heat J/kg.K , $[\Delta T/\Delta t]$ is rate of change of temperature, $[dT/dx]$ is temperature gradient $[(T_2 - T_5)/(x_5 - x_2)]$

Procedure:

1. Insert the heating element into open end of the hollow metal rod. Heat the rod for about half an hour so as to attain steady state.
2. Connect the output of the thermometers to the temperature and timer module.
3. Note down the temperature at different distances x as in Table 1. Plot a graph T vs X .
4. Find the ratio $[(T_2 - T_5)/(x_5 - x_2)]$ from the graph.
5. Now turn off the heating. Reset the timer. Note down the temperature of all six thermometers at an interval of two minutes, up to ten minutes as in Table 2.
6. Calculate $\Delta T = [T @ 0s \sim T @ 600s]$ and consider $\Delta t = 600$ s. Compute the sum $\sum_0^L \frac{\Delta T}{\Delta t}$ as indicated in Table 1 and Table 2.

Table 1:

Time t (sec)	Temperature $^{\circ}\text{C}$					
	T_1 (at x_1)	T_2 (at x_2)	T_3 (at x_3)	T_4 (at x_4)	T_5 (at x_5)	T_6 (at x_6)
0						
120						
240						
360						
480						
600						
$[\Delta T/\Delta t] =$						



Table 2:

x, cm	Δx , m	$\left[\frac{\Delta T}{\Delta t} \right]$	$\left[\frac{\Delta T}{\Delta t} \right] \Delta x$
5	0.05		
10	0.05		
15	0.05		
20	0.05		
25	0.05		
30	0.05		
$\sum_0^L \frac{\Delta T}{\Delta t} \Delta x$		Total	

Graph:



Observations:

Density of the material (brass or iron) of the rod, $\rho = (8520 \text{ or } 7850) \text{ kg/m}^3$ Specific heat of the material (brass or iron) of the rod $s = (401.93 \text{ or } 452) \text{ J/kg.K}$

Substitution & Calculation:

- $\sum_0^L \frac{\Delta T}{\Delta t} \Delta x$ from the table =
- dT/dx , the temperature gradient $[(T_2 - T_5)/(x_5 - x_2)]$ from the graph =
- $k = \frac{\rho s \sum_0^L \frac{\Delta T}{\Delta t} \Delta x}{\left(\frac{\Delta T}{\Delta x} \right)_{x_2 \sim x_5}} = \text{--- -- W/mK}$

Result: The thermal conductivity of the given good conductor is, $k = \text{--- -- W/mK}$

**Problems:**

Formula	
Quantity of heat energy, Q	$Q = \frac{\kappa A (T_1 - T_2) t}{x} \text{ (J)}$
Thermal conductivity, κ	$\kappa = \frac{Qx}{A (T_1 - T_2) t} \text{ (W/m/K)}$
Lorentz number, L	$L = \frac{\kappa}{\sigma T} \text{ (W}\Omega\text{K}^{-2}\text{)}$
Thermal conductivity, κ	$\kappa = \frac{1}{2} n v k_B \lambda \text{ (W/m/K)}$
Thermal conductivity, κ	$\kappa = \frac{1}{3} C_e v \lambda \text{ (W/m/K)}$
Electrical conductivity, σ	$\sigma = \frac{ne^2\tau}{m}$



P1. Calculate the heat conducted per hour through a concrete ceiling with coefficient of thermal conductivity 0.18 W/m/K. The area is 1000 m², thickness is 1.5 m, and the temperature on either side of the ceiling is 30 °C and 60 °C

Soln:

$$Q = \frac{\kappa A (T_1 - T_2) t}{x} \text{ (J)}$$

$$Q = \frac{0.18 \times 1000 (303 - 333) 3600}{1.5} \text{ (J)}$$

$$Q = 1.3 \times 10^7 \text{ J}$$

P2. A thermally insulated copper rod of 19 cm length and 0.785 cm² of cross section is heated at one end through 100 °C while the other end is kept at 30 °C. If the amount of heat flowing through the rod in 10 minutes is 6.6 kJ, calculate the thermal conductivity of copper.

Soln:

$$\kappa = \frac{Qx}{A (T_1 - T_2) t}$$

$$\kappa = \frac{6.6 \times 10^3 \times 19 \times 10^{-2}}{0.785 \times 10^{-4} (100 - 30) 600}$$

$$\kappa = 380.3 \text{ (W/m/K)}$$

P3. Calculate the electrical conductivity and Lorentz number of a metal at 300 K with the relaxation time 10⁻¹⁴ s and thermal conductivity 123.9 (W/m/K) and free electron concentration 6 × 10²⁸ m⁻³

Soln:

$$\sigma = \frac{ne^2\tau}{m}$$

$$\sigma = 1.68 \times 10^7 / \Omega m$$

$$L = \frac{\kappa}{\sigma T} = \frac{123.9}{1.68 \times 10^7 \times 300}$$

$$L = 2.447 \times 10^{-8} \text{ (W}\Omega\text{K}^{-2}\text{)}$$

P4. Calculate the thermal conductivity of a metal from classical expression having free electron concentration 8.5 × 10²⁸ m⁻³ and mean free path 4 × 10⁻⁸ m at 27 °C.

Soln:

$$\kappa = \frac{1}{2} nv k_B$$

to determine, v

$$E_F = \frac{1}{2} mv^2 \Rightarrow v = \left(\frac{2E_F}{m} \right)^{\frac{1}{2}}$$

Wkt, $E_F = k_B T$

$$\therefore v = \left(\frac{2k_B T}{m} \right)^{\frac{1}{2}} = 95335 \text{ m/s}$$

thermal conductivity is

$$\kappa = \frac{1}{2} nv k_B$$

$$\kappa = 2236 \text{ (W/m/K)}$$

**Question Bank:**

Question	Marks (Approximate)
Mention any four assumptions of classical free electron theory.	4
Discuss the failures of classical free electron theory.	6
Explain in detail, any three merits of quantum free electron theory.	6
Define Fermi function. Explain the variation of Fermi function with energy and temperature.	8
Mention the postulates of quantum free electron theory and explain any two of its merits.	8
Derive the expression for Fermi energy at $T=0K$.	5
Find the temperature at which there is 1% probability that a state with energy 0.5 eV above Fermi energy will be occupied.	4
Calculate the electrical conductivity for a metal with relaxation time 10^{-14} s at 300 K.	4
Calculate the Fermi energy in eV, Fermi velocity and Fermi temperature for sodium, given the density is 971 kg/m^3 , atomic weight is 23 and it has one conduction electron per atom.	4
Calculate the probability of an electron occupying an energy level 0.02 eV above and below the Fermi level at 200K	4